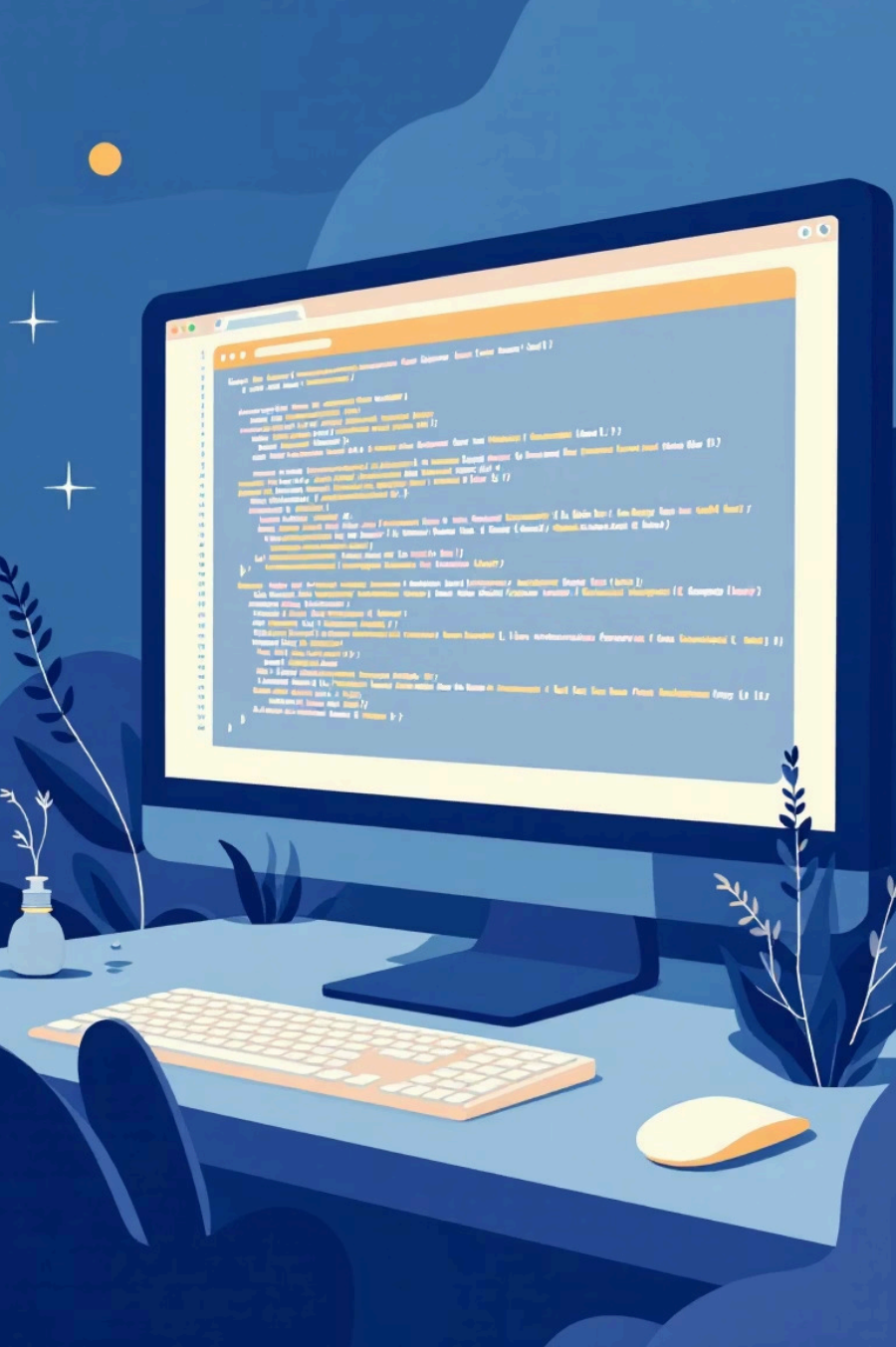




Introduction to Programming

Programming is the process of writing instructions that a computer can understand and execute to perform specific tasks. These instructions are written using a programming language, which defines rules (syntax) and structure that the computer follows.

Compilation Types

AOT (Ahead Of Time)

The code is compiled before the application runs. This gives better performance and faster execution.

JIT (Just In Time)

The code is compiled while the application is running. This helps during development because it allows hot reload and faster testing.

What is a Programming Language?

A programming language is a way for humans to communicate with computers. It provides:

- Syntax: rules for writing code
- Structure: how instructions are organized
- Semantics: meaning of the instructions

Introduction to Dart Language

Dart is an open-source, client-optimized programming language developed by Google. It is mainly used with Flutter to build modern applications.

Why Dart?

- Easy to learn: Similar to Java, C++, and JavaScript
- Fast and efficient: Compiles to native code
- Object-Oriented: Supports classes, objects, inheritance, and abstraction



Google

Made with GAMMA

Where is Dart Used?



Flutter Development

Mobile, Web, Desktop applications



Backend Services

Using frameworks like Shelf



Command Line Tools

Automation and utilities

Dart Basics: Variables & Data Types

A variable is a container used to store data in memory.

Variable Keywords

Variables are used to store values in memory. Dart provides different keywords to control how variables behave.

- **var:** Dart automatically detects the data type at compile time. Once assigned, the type cannot change.
- **final:** The value can be assigned only once. It is decided at runtime and cannot be changed later.
- **const:** The value must be known at compile time and can never change.

```
var name = "Rana"; // Type is String, cannot change later  
final age = 25; // Assigned once at runtime  
const pi = 3.14; // Compile-time constant
```

Common Data Types

```
int age = 25; // Whole numbers
```

```
double price = 10.5; // Decimal numbers
```

```
String name = "Rana"; // Text
```

```
bool isActive = true; // True or False
```

Dynamic Type

The dynamic keyword allows a variable to change its type during runtime.

```
dynamic value = 10;  
value = "Hello";  
value = true;
```

Important Notes

- Dart is strongly typed
- Using dynamic removes compile-time type checking
- Errors may appear at runtime



Collections in Dart

Collections are used to store multiple values in one variable.

Lists in Dart

A List is an ordered collection of elements where each item has an index (starting from 0).

```
List<int> numbers = [1, 2, 3];
```

Adding Elements

```
numbers.add(4); // Adds an element at the end of the list
numbers.insert(1, 10); // Inserts 10 at index 1 and shifts other elements
numbers.addAll([5, 6]); // Adds multiple elements at once
```

Removing Elements

```
numbers.remove(2); // Removes the first occurrence of value 2
numbers.removeAt(0); // Removes the element at index 0
numbers.clear(); // Removes all elements from the list
```

Accessing List Data

```
print(numbers[0]); // Gets the first element
print(numbers.length); // Gets the number of elements in the list
```

Dynamic List

```
List<dynamic> data = [1, "Hello", true]; // List can store different data types
```